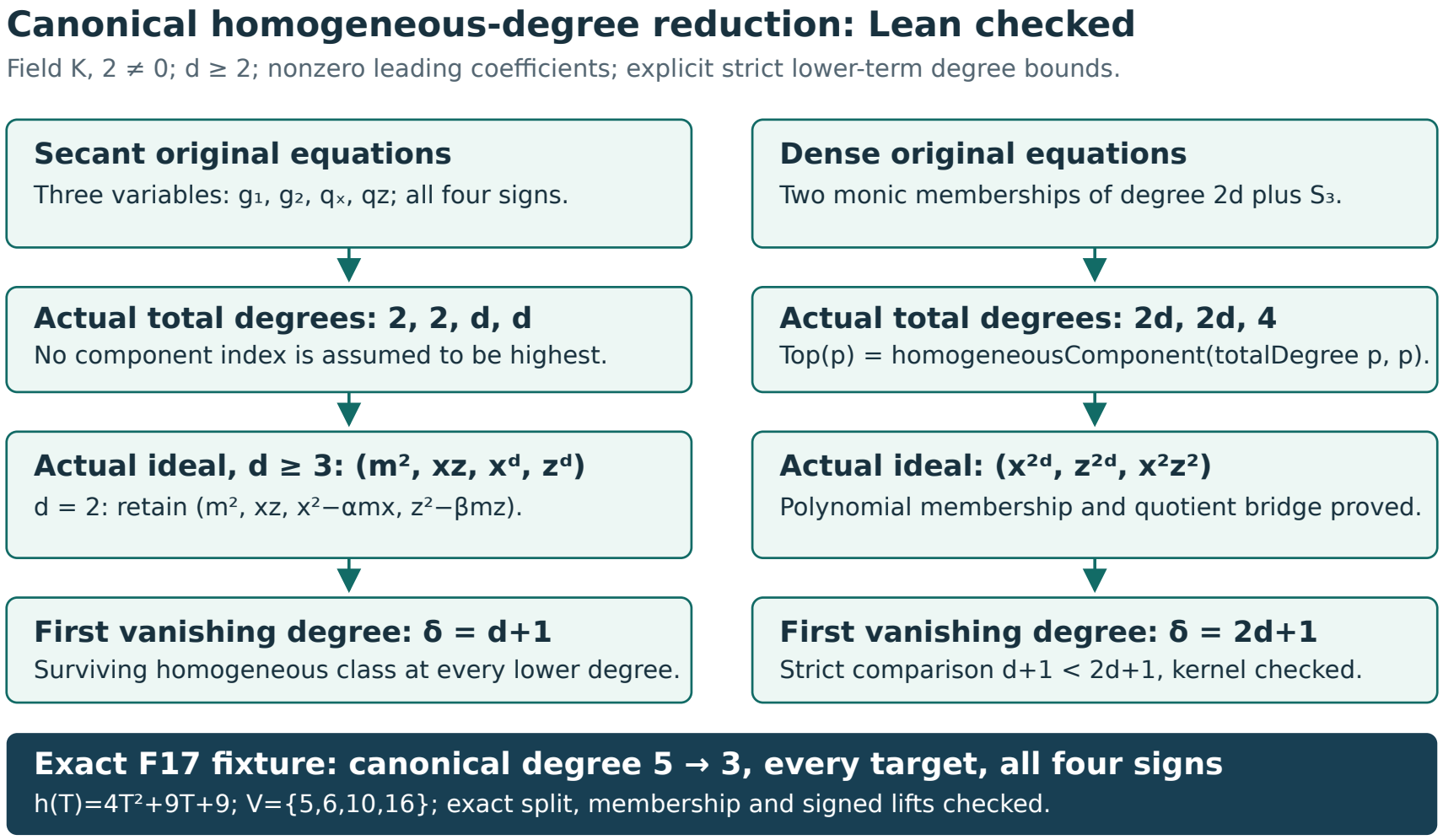
Lean proof of the canonical top-ideal degree reduction

2026-10-31 · prime fields · two summands · formalization following PR 3077

The canonical highest-total-degree ideals of the displayed secant and dense systems have first vanishing homogeneous degrees d+1 and 2d+1 for d≥2, under the explicit field/coefficient/lower-degree hypotheses. The F17 fixture is checked at 5→3 for all targets and all four signs. Each original equation's actual total degree is proved; the statement does not assume which component is highest.



Headline declarations: DegreeBridge.canonical_strict_reduction and CanonicalFixture.degree_five_to_three. Full proof map: FORMALIZATION.md.

Exact scope and verification

Invariant: top(p) is the homogeneous component at p.totalDegree. The actual ideal is the span of these tops. FirstVanishing J n states that every degree-n homogeneous polynomial belongs to J and that, at each smaller degree, some homogeneous polynomial does not. The quotient-map equivalence is proved in Lean.

Hypotheses: K is a field, $2 \neq 0$, $d \geq 2$, and both graph leading coefficients are nonzero. Graph lower terms have total degree below d; the dense monic membership lower terms have total degree below 2d. Targets and coefficients r,s,A,B are arbitrary for the degree statement. Exact group-relation meaning uses the separate geometric hypotheses in the earlier written proof.

Canonical highest forms	DegreeBridge proves the actual total degree of every generator and equality of canonical and computed top ideals.
Higher degrees	Mathlib’s exact monomial-ideal support criterion connects the exponent lemmas to actual ideal membership and homogeneous quotient images.
Quadratic case	The ideal remains nonmonomial. All cubics vanish; a map into an iterated square-zero extension proves that mx survives in degree two.
Concrete F17 base	Exact split and full intersection, cardinality 4, nonsingularity, signed curve lifts, actual membership product and four-branch canonical degree 5→3 are checked.
Exact secant algebra	The cubic identity, residual-to-factorization implication and selected-point curve membership are checked over any commutative ring.
Kernel checks	Eleven modules build with warnings treated as failures. The unchanged transitive auditor checks 121 declarations, including generated equations:0 failures,0 native trust.
Remaining formal scope	Full elliptic-curve point addition/group-equivalence is written, not formalized. No Hilbert-series coefficient, F4-degree or runtime theorem is inferred.

Foundations and pins: Lean 4.19.0; Mathlib v4.19.0, commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. Dependencies are locked. All transitive axioms are confined to propext, Classical.choice, Quot.sound. There are no placeholders, custom axioms or native-evaluation exceptions. The cache returned proxy CONNECT 403, so the pinned dependency was built from source. No network-policy bypass was attempted. lean4checker was unavailable; no fresh-checker result or passing cache-download gate is claimed.

Reproduce: commands and archive, build, axiom audit, source hashes and environment. Failed development logs remain in a deterministic archive, separate from the final successful receipts.

Provenance: field/curve parameters match `icv1-fp5-tm1-a77cc6af`, candidate FB1hed5b19904cc8. The Lean files check polynomial data, not storage-hash encoding or group order. Sources: parent SECANT_GRAPH/THEORY; Mathlib ideal/homogeneous theory square-zero extension. Literature-wide novelty remains open. This proof-only continuation adds no cryptographic experiment, CAS rerun or cost/leaderboard claim.